Practice Quiz: Communication / Interdisciplinary Methods (Research Methods)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. Active Inference's shared formal language across disciplines is: A) English B) The mathematics of variational free energy — providing a common framework that translates between domain-specific concepts C) Computer code D) Clinical terminology

2. The most common misunderstanding about Active Inference is: A) It's too simple B) That the brain literally calculates Bayes' theorem — when in fact Active Inference is a computational-level theory that doesn't prescribe specific neural implementations C) It's only for robots D) It's purely philosophical

3. Pre-registration in computational modeling studies: A) Is unnecessary B) Prevents post-hoc model selection bias by committing to hypotheses, models, and analysis plans before data collection C) Replaces data analysis D) Is only for clinical trials

4. Open-source software benefits Active Inference by: A) Limiting access B) Enabling replication, community development, standardization of methods, and collaborative improvement C) Reducing quality D) Increasing cost

5. When communicating to clinicians, Active Inference should emphasize: A) Mathematical proofs B) Clinical utility — patient outcomes, treatment implications, and diagnostic precision C) Physics analogies D) Robot implementations

6. Multi-site Active Inference studies require: A) Identical labs B) Standardized tasks, models, and analysis protocols — enabling pooling across sites while maintaining local flexibility C) Single investigators D) No coordination

7. Marr's three levels (computational, algorithmic, implementational) help communicate Active Inference because: A) They constrain the framework to one level B) They clarify that

Active Inference primarily operates at the computational level (what problem is solved) — algorithmic and implementational details can vary, preventing confusion with neural mechanism claims C) Only the implementational level matters D) All three levels say the same thing

8. Reproducibility challenges unique to computational modeling studies include: A) Only data sharing issues B) Sensitivity to optimization settings, model specification choices, prior sensitivity, and software version dependencies — requiring detailed reporting of all model fitting parameters and random seeds C) Only statistical thresholds D) Only sample size concerns

Part B: Short Answer

1. You are presenting Active Inference to an audience of clinical psychologists with no mathematical background. Describe how you would explain "precision weighting" using a concrete clinical example, without any equations. (200 words)

2. A reviewer criticizes your Active Inference paper for being "unfalsifiable" — claiming that the framework can explain any result post-hoc. Write a brief response defending the falsifiability of Active Inference by identifying two specific empirical predictions that would disconfirm the theory. (200 words)

Part C: Essay Questions

1. Design a complete interdisciplinary research program (3-5 years) applying Active Inference to a real-world problem (climate change adaptation, educational reform, public health). Specify: (a) the problem and why Active Inference helps, (b) the disciplinary team needed, (c) 3-4 specific aims, (d) methods for each aim, (e) expected outcomes, (f) broader impacts. (600 words)

2. Write two versions of the same 300-word abstract for a research paper on Active Inference and autism: one for *Biological Psychiatry* (clinical audience) and one for *PLoS Computational Biology* (computational audience). Reflect on what you emphasized differently and why.

3. Develop an ethics protocol for a computational psychiatry study using Active Inference. Address: (a) informed consent for computational analysis, (b) data privacy for computational phenotypes, (c) incidental findings (discovering potential psychiatric vulnerability through computational parameters), (d) equitable access to computational assessments, (e) cultural sensitivity in applying Western computational models cross-culturally. (500 words)